

with the surface affine connection given by

$$\gamma_{ab}^c = -\hat{v}^c \partial_a \tau_b + \frac{1}{2} h^{cd} (\partial_a \bar{h}_{bd} + \partial_b \bar{h}_{ad} - \partial_d \bar{h}_{ab}) \quad (45)$$

$$= \Gamma_{ab}^c + u_\mu^c \partial_a u_b^\mu = u_\mu^c u_a^\nu \nabla_\nu u_b^\mu, \quad (46)$$

and associated torsion tensor F_{ab} defined through

$$2\gamma_{[ab]}^c = -\hat{v}^c F_{ab}. \quad (47)$$

A submanifold Riemann tensor $\mathcal{R}_{bcd}^a$ can be constructed from γ_{ab}^a through the standard form

$$\mathcal{R}_{bcd}^a = \partial_c \gamma_{db}^a - \partial_d \gamma_{cb}^a + \gamma_{cf}^a \gamma_{db}^f - \gamma_{df}^a \gamma_{cb}^f. \quad (48)$$

The boost invariant extrinsic curvature of the submanifold K_{ab}^I can be defined from the covariant derivative of the tangential vector through

$$K_{ab}^I = -u_a^\mu u_b^\nu \nabla_{(\mu} n_{\nu)}^I = n_\mu^I D_a u_b^\mu + \frac{1}{2} \hat{v}^I \tau_{ab}, \quad (49)$$

and transforming under mass U(1) transformations as

$$\delta K_{ab}^I = \frac{1}{2} F_{Ia} \partial_b \sigma + \frac{1}{2} F_{Ib} \partial_a \sigma. \quad (50)$$

Notice that the covariant derivative D does not act on normal indices I . It is also possible to define a covariant derivative $\mathcal{D}$ that acts on all types of indices. In particular we will find that

$$\mathcal{D}_a T^I = D_a T^I - \omega_a^I{}_J T^J. \quad (51)$$

The tensor $\omega_a^I{}_J$ is called the external rotation tensor and can be interpreted as a SO(d-p) connection, it is defined from the covariant derivative of the normal 1-form

$$\omega_a^I{}_J = n_J^\mu D_a n_\mu^I, \quad (52)$$

it is antisymmetric in I, J indices, and transforms under U(1) mass transformations as

$$\delta \omega_a^I{}_J = -\frac{1}{2} (F_{aJ} \partial^J \sigma + F^I{}_J \partial_a \sigma + \tau_a^I \partial_J \sigma). \quad (53)$$